

Homogeneous finite metric spaces of small orders

June 6, 2026

Let X be a metric space of size $n < \infty$.

Note that for isometries of finite metric spaces, the only importance is what distances are equal, and all possibilities are valid (to see this, note that if $d_i = 1 + 1/i$ then $d_1, \dots, d_n$ satisfy $d_i \leq d_j + d_k$ and so are valid pairwise disjoint distances). Thus the problem of if a finite metric space is completely inhomogeneous, homogeneous, strongly homogeneous or perfectly homogeneous can be reduced to the coloring of its complete graphs in such a way, that each color corresponds to a different distance. Then an isometry becomes a map which preserves the colorings between vertices of the graph.

As such, I will call a 3-point space a "triangle" since any such space corresponds to vertices of (possibly degenerate) triangle, even though an actual triangle is an infinite metric space. In the same way I might refer to a 4-point space as a parallelogram, or rhombus, depending on the coloring of its complete graph.

For metric spaces the following implications hold

$$\text{perfectly homogeneous} \implies \text{strongly homogeneous} \implies \text{homogeneous}$$

where **homogeneous** means for any two points $a, b \in X$ there is an isometry $f : X \rightarrow X$ with $f(a) = b$, **strongly homogeneous** means that for any two isometric $A, B \subseteq X$ there is an isometry $f : X \rightarrow X$ with $f(A) = B$, and **perfectly homogeneous** means for any isometry $g : A \rightarrow B$ where $A, B \subseteq X$ there is an isometry $f : X \rightarrow X$ extending g . A space is **completely inhomogeneous** where there is a unique isometry $f : X \rightarrow X$, namely the identity map.

Cases $n = 0, 1$ correspond to the *empty space* and *a point*. They are the only metric spaces which are both homogeneous and completely inhomogeneous.

Case $n = 2$ corresponds to a *segment* which is perfectly homogeneous.

Case $n = 3$ leads to three different possible colorings, *equilateral triangle* which is perfectly homogeneous, *isosceles triangle* which is inhomogeneous but not completely inhomogeneous, and *triangle with different sides* which is completely inhomogeneous.

For $n \geq 4$ there starts to be much more colorings. Only a few of them are homogeneous. A necessary condition for a coloring of a complete graph to correspond to a homogeneous metric space, we need that there is the same

amount of edges of each color coming out from each vertex, which reduces the cases significantly.

For $n = 4$ there is three homogeneous colorings, *regular tetrahedron*, *a square* and a *rectangle*, all of which are perfectly homogeneous.

For $n = 5$ there is two homogeneous colorings, one using one color, second using two colors, both are perfectly homogeneous.